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MSc Computer Science with Artificial Intelligence

Impact of Temporal Structure and Duration of Actigraphy on ADHD Prediction with Fair and Interpretable Machine Learning

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Executive Summary

Attention-Deficit Hyperactivity Disorder (ADHD) prediction has the potential to unlock solutions to the UK's under-diagnosis crisis; realising health benefits and social opportunities for those stuck on waiting lists and saving the taxpayer estimated £billions. Novel opportunities to validate methods for collecting ADHD data from wearable health trackers and apply ethical artificial intelligence to resolve assessment delays motivates this study. Current research shows promise but a gap exists to test whether biomarkers from activity data – intra-daily (RQ1) and weekly (RQ2) temporal factors – can predict ADHD beyond the small convenience samples and opaque modelling approaches currently limiting transition of prediction technologies into clinical practice. This study addresses these limitations with interpretable, fair modelling on large-cohort MCS data, representative of UK diversity. Further, the study reports on optimal activity monitoring durations (RQ3), minimising data collection costs while retaining reliable results, key for real-world application.

A quantitative, post-positivist approach on secondary MCS data processed age-7 children actigraphy and parent-questionnaire responses into a valid sample (N=5,033) for machine learning. Predictor variables were extracted automatically from minute-to-minute timeseries data using tsfresh and handcrafted from daily aggregates to explore intra-daily patterns (RQ1). Data were then weekend-stratified to compare weekly structure using Wilcoxon signed-rank significance testing (RQ2). A range of feature selection methods and model comparison between Logistic Regression, Naïve Bayes and Extra Trees developed a validated machine learning pipeline for reproducible analysis of covariates, subgroups and monitoring durations (2-6 days), the latter applied Holm-adjusted bootstrap resampling for significance testing RQ3.

The tsfresh extracted features proved consistently superior to the handcrafted daily aggregates, suggesting intra-daily activity patterns are important for differentiating ADHD (RQ1). Likewise, conditioning the data on weekdays/weekends improves AUROC significantly (+0.093, $p < 0.0001$), evidencing weekly structure present in large cohorts (RQ2). L2-regularised Logistic Regression proves the most interpretable and performative (AUROC 0.846) on the selected weekend-stratified tsfresh features, with feature importance supporting construct validity. Covariate analysis reveals contributions from Sex, Ethnicity and parent-questionnaire symptom data improve performance but cautions on bias-free implementation. Further subgroup analysis demonstrates mild discrimination between sexes and model miscalibration, likely influenced by

low class prevalence. 5-day monitoring duration is found to be comparable with the 6-day reference, while 2 and 3-day durations significantly underperform ($p < 0.001$).

The high ethical standards required for supporting healthcare influenced this study's focus on interpretable and equitable methods, in accordance with the TRIPOD+AI framework for transparency in clinical prediction modelling. This study recommends using automatic feature extraction on raw actigraphy in order to exploit intra-daily patterns while remaining interpretable through coefficient analysis. Stratifying weekends is also advised for capturing ADHD differences across the week. Large, diverse samples from reputable sources, such as MCS, are shown to aid analysis of inequalities/bias, although findings implore for even more data and improved understanding of covariate causality. The study contributes novel findings, recommending at least 5 days activity monitoring for reliable ADHD prediction.

1. Introduction

1.1 Background

ADHD is a neurogenetic disorder [2], affecting 5.6% of the UK population [3]. A complex set of symptoms including hyperactivity, inattention and impulsivity cause considerable day-to-day challenges for those affected by the disorder: disruptive behaviours lead to school exclusions and family conflict, risk of driving accidents, crime and employment instability increase and secondary disorders develop such as addiction and substance misuse, anxiety, low self-esteem and depression – often mistreated without confirmation of underlying ADHD [2], [4], [5]. The costs to society are also high, unsupported ADHD has been estimated at £17 billion annually to cover long-term unemployment, justice system and healthcare costs [5]. And yet these costs to the individual and their communities are avoidable with acclaimed researchers and *National Institute for Health and Care Excellence* (NICE) guidance explicitly linking diagnosis and the interventions that follow to improved access to education and employment and positive health outcomes, with pharmaceutical intervention effective in over 90% of cases [2], [6], [7].

A surge in demand for ADHD assessment, driven by improvements to restrictive criteria and better community recognition [8], has beleaguered an already strained healthcare system unable to supply enough specialists, causing extreme treatment delays up to 15 years [4], [5], [9]. An independent ADHD taskforce reports an under-diagnosis crisis and, in-line with government recommendations [10], supports technology-driven solutions to aid understaffing and assessment pathway delays [5]. ADHD prediction research seeks to facilitate assessment with objective and reliable modelling able to differentiate the condition based on data captured from biosensors such as fitness trackers [11]. The current state of research has shown promising progress [12], [13], [14], but adopting these technologies into practice requires more work to standardise data processing approaches and align with ethical and accountable regulations and legislation [15], [16], [17], [18]. The *Transparent Reporting of a multivariable prediction model for Individual Prognosis Or Diagnosis + Artificial Intelligence* (TRIPOD+AI) statement provides a framework for meeting these standards, promoting transparent and fair clinical prediction modelling and reporting [19].

A theoretical framework for how to model activity data for predicting ADHD has been well-established over the last 20 years [12], [14]. However, generalising theory into practice is

hampered by small convenience samples, limited transparency and a lack of methods for examining fair processing [13], [20], [21], which can exacerbate inequalities such as reduced access to care for females and ethnic minorities [22], [23]. Significant factors for modelling, in particular the temporal structure of how and when symptoms present [2], [24], [25], need to be validated on larger cohorts representative of diverse populations. In addition, research in this area has yet to study the effects of actigraphy duration on ADHD prediction, a result that could advise on optimal data collection times as part of a standardised approach for supporting clinical assessment in a way that is sensitive to the children being assessed while delivering reliable data toward a diagnosis.

A suitable large-cohort data source has been identified for this study, the MCS collected actigraphy on children age 7 as part of longitudinal research on a UK-wide sample of nearly 19,000 people with well-stratified demographic representation [26].

1.2 Aims & Objectives

Aims

Investigate the effect of temporal differences and monitoring period duration on ADHD prediction performance on a large-cohort. The study is UK-focused using secondary data from MCS to develop a machine learning pipeline for comparing the various temporal conditions while demonstrating a modelling approach acceptable for real-world, clinical application.

Objectives

- Sample the MCS data for valid actigraphy, class labels and the covariates necessary for equitable modelling, process the activity data into ADHD predictor variables
- Develop the machine learning pipeline on complete duration data for assessing performance and selecting for optimal predictor variables and learning algorithms conducive to study aims, including analysis of temporal differences
- Validate the pipeline, selected variables and model with interpretable methods and sensitivity analysis for subgroup discrimination
- Re-train prediction models on reduced duration data conditions on the validated pipeline and assess the effect on performance with statistical testing

Research Questions

RQ1. Can extracting intra-daily activity features improve on daily aggregate features for predicting ADHD using cross-validated Area Under the Receiver Operating Curve (AUROC) on large-cohort MCS data?

RQ2. Does stratifying weekday and weekend activity features significantly improve on non-stratified equivalent for predicting ADHD using cross-validated AUROC on MCS data?

RQ3. How do 2 – 5-day monitoring period durations affect cross-validated AUROC ranking and stability for predicting ADHD compared with a 6-day reference duration?

2. Literature Review

2.1 ADHD

In response to the under-diagnosis crisis, NHS and government both advocate for technology-driven solutions to fill the gap by guiding less experienced practitioners toward reliable assessment and expediting diagnosis by those already qualified [5], [10]. However, technology can only support clinical decision making if it can faithfully replicate the expert's role [21], including reliable methods for data collection and processing [27].

Modelling expert processes reliably depends on understanding the complexity of the condition in terms of its scope and assessment criteria [28]. ADHD is known to be heterogenous across sub-groups, sub-types and temporal factors in terms of how symptoms present [2], [24]. Any analysis dependent on symptoms will need to account for these confounding differences to conserve accuracy and fairness, for example treating male and female symptoms as equivalent undermines a typically internalised female presentation [24], leading to sub-group underdiagnosis [7], [29].

Prevalence and assessment

Historically, the UK has had a more restrictive assessment criteria (*International Classification of Diseases*, ICD) than its US counterpart (*Diagnostic and Statistical Manual of Mental Disorders*, DSM) [2], [4], [30], [31], seen consistently in UK survey data giving lower prevalence in 1999, 2004 and 2017 of around 1.5 - 2% in adolescents [8], [32]. Recent National Health Service (NHS) reports [4], now cite globally recognised prevalence rates closer to 5% [2], [3], likely due to increased awareness and closer alignment of the latest ICD-11 with DSM-5 criteria [4], both now accepted for UK diagnosis [7]. This materialises as chronic underdiagnosis in the UK over prior decades [5], now realised as the sudden influx in demand for assessment and treatment this study seeks to address.

Defining the condition is challenging, with debate ongoing regarding the nature of the symptoms and the degree to which these must deviate from 'normal' [2]. As a result assessment criteria changes over time [33], with current methods relying on a combination of rating scales, reported impairments and contextual observations to meet clinical criteria [7].

However, Barkley argues common traits such as emotional dysregulation and sleep disturbances are overlooked, and that continuously distributed symptoms are forced into categories for clinical convenience [2]. The UK's Royal College for Psychiatrists concur, suggesting future assessment methods should adapt [34].

Heterogenous presentation

ADHD presents differently across sub-groups for age, sex and ethnicity [5], [8]. Prevalence decreases with age as symptoms generally become less pronounced [3], [35] and the condition affects three times as many boys as it does girls who, unlike boys, are predominantly of the inattentive subtype [24]. These differences result in inequalities such as increased missed or delayed diagnosis for females, the socially disadvantaged and ethnic minorities [5], [23], [36]. Yet, these differences are only partially accounted for in the literature and where not considered are likely to have confounding effects [12], [13]. Exceptions include Rahman who stratified subgroups across ADHD and non-ADHD classes [37], and Brandt et al who controlled for these demographics at the outset [32], yet neither assess their *results* for inherent inequalities, which others recommend [38].

Socio-economic status (SES) is also associated with ADHD [8], although the causal connection has yet to be established [2]. Some studies include SES as a variable for predicting ADHD [32], [37] yet this may lead to algorithmic bias – discussed further in 2.3 – for example, low-income should not be a factor for diagnosis.

Temporal context is important, high activity levels during school lunch-break are normal, whereas similar levels in the classroom constitute deviation [2], and consistency of symptoms over time and place is required criteria [7]. Circadian factors are well-established in ADHD [39], with prediction studies often dependent on inter- and intra-daily variances [40], [41], [42] or stratifying day/night [37], [43]. In addition, Pedersen et al find symptom's worsen during school compared to home/weekends, in particular in the afternoon [25], consistent with term-time/holiday and Summer/Winter differences found elsewhere [44], [45], [46].

These ontological and heterogenous challenges present an opportunity for machine learning, where hard to define concepts are learned by example, and continuous variables are naturally

accommodated [47], [48]. Predictive models can then make objective inference on new examples, effectively using combined expertise to make accurate and consistent predictions for aiding diagnosis. The next section looks at how autonomous biosensing may accurately and reliably capture the observations needed to predict ADHD with machine learning, paving the way for technology-based solutions to the under-diagnosis crisis and providing a framework for analysing reliability in the data collection process.

2.2 Digital Phenotyping

ADHD assessments rely on experts such as specialised paediatricians and psychiatrists [5] to spend time observing children for signs of behaviour symptomatic of the condition [30], [31], a manual and time-consuming approach potentially prone to the subjectivity of the assessor [12]. Digital phenotyping offers a method to capture those observations objectively and autonomously through biosensors such as wearable devices and neuroimaging for inferring an individual's health [49]. For example, Brandt et al demonstrate ADHD correlates with reduced sedentary activity, measured with accelerometer data calibrated to various activity levels (actigraphy) [32], a finding corroborated by others [37], [50], [51].

Translating the raw data from biosensors into valid and reliable biomarkers – derived variables that represent a condition's distinguishing features or *phenotypes*, such as sedentary activity – enables inference through statistical methods [52], and more advanced machine learning approaches for handling large datasets [47], that may include multiple biomarkers along with questionnaire data for known covariates such as gender and ethnicity [4], [24].

Constructing valid biomarkers

Actigraphy biomarkers for ADHD have been studied for decades, Crescenzo et al round up 24 prominent early studies, finding that derived variables for mean daily activity levels and sleep patterns measured over 7-14 days are consistently associated with the condition [14]. Recent studies explore more advanced techniques such as functional linear modelling to investigate circadian rhythms [41], non-linear parameters such as kurtosis and skewness of the actigraph [53], and automated feature extraction [54], all able to produce interpretable features and predict the condition, although not always improving on the well-established biomarkers above

and with methods still in pilot/early phase of research yet to be repeated and verified by others.

Actigraphy boasts a number of benefits, including the low cost, ecological and non-invasive nature of the monitoring devices [12], [55]. However, Basic et al argue in their critical review of ADHD motion sensing that activity alone is insufficient to capture the full range of symptoms, especially across different sub-types [12]. While there is evidence that actigraphy *can* differentiate sub-types [41], [42], [56], their recommendation toward multi-sensor approaches is logical, for example heart-rate variability (HRV) captures dysregulated autonomic nervous system activity, a known ADHD symptom [57], distinct from physical activity.

The HYPERAKTIV study demonstrates multi-sensor capture with 7 days of actigraphy combined with 21 hours HRV and questionnaire responses [54], tsfresh is applied to each sensor's raw timeseries to automatically extract large, generic sets of derived variables including means, variances, skewness and kurtosis [58]. Mutual information then determines a reduced feature set able to predict ADHD (AUROC 0.76). Unlike the handcrafted features mentioned earlier [12], [14], these are not based on prior theory. However, they allow unknown aspects of the feature space to be explored, produce predictive power and remain interpretable [58].

Kaur and Kahlon working with the same dataset use principal component analysis (PCA) as a means to reduce the automated features and report substantial accuracy > 0.99 [59]. However, PCA reduces interpretability [60], while opaque methodology and lack of hold-out testing limit validity due to potential overfitting [48], or temporal leakage of medication-related variables [61]. In contrast, Rahman presents reproducible and interpretable methods for analysing Fitbit data (activity/HRV) yielding more reliable results (AUROC, F1 c.0.9) [37].

The broad manifestation of ADHD symptoms, clearly challenging to capture and represent, develop out of more fundamental, neurological aetiology such as executive function deficits [28]. Electroencephalography (EEG) and functional Magnetic Resonance Imaging (fMRI) offer methods for capturing these *direct* biomarkers [52], with impressive predictions appearing to validate the efficacy of this approach with AUROC and F1 scores > 0.9 [62], [63], [64], [65]. However, Pulini et al critically review 69 neuroimaging studies and find over 15% lack methodological detail and/or rely on circular analysis leading to bias and inflated accuracies, while the most rigorous research peak around AUROC 0.8 [20], suggesting no clear advantage over wearables in terms of predictive power. Moreover, neuroimaging tends to be lab-based

and more intrusive than wearable devices, compromising ecological validity and ethical acceptability [13], [55], and the methods for processing raw image data are typically non-interpretable – based on neural architectures [48], incongruent with accountable decision making [5].

Monitoring period reliability

Children’s physical activity is known to vary significantly within and between days [66], and ADHD symptoms in particular are inconsistent over time [25]. Reliability therefore depends on accounting for this variation by monitoring over a long enough duration to obtain consistent, repeatable data [67]. For example, a child with ADHD whose symptoms are less prevalent on a given day may be misclassified until further days are included.

While significant work has gone into verifying minimum wear-times for physical activity [27], reliability of ADHD biomarkers derived from physical activity has yet to be investigated. A range of monitoring periods have been shown to predict ADHD, from 3 hours to 3 weeks for naturalistic settings [37], [40], [68]. Yet attempting to compare durations across studies is limited due to heterogenous study designs [12], [13].

Facilitating healthcare diagnoses of ADHD with digital phenotyping, therefore, relies on applying correct methods for processing sensor data into valid biomarkers and modelling the covariates of the condition to account for variation across gender, age, ethnicity and temporal factors, such as school/home differences. Yet, accounting for all these conditions is seldom seen across the literature, gender and age are sometimes considered [12], [13], and occasionally weekends are controlled for [37], [41], but most studies rely on small, clustered samples [13], [20] and simply do not have the data to offer generalisable results. Large-cohort studies offer an exception [26], [69], [70], although no study has yet determined the effect of monitoring period duration on ADHD prediction performance, essential for translating findings into real-world recommendations and developing externally valid models for clinical practice [5].

2.3 Machine Learning for ADHD prediction

Once biomarkers have been derived from the raw data, they can be analysed using a variety of statistical methods to make inferences and predictions [52]. Brandt et al, like many actigraphy studies before [12], [14], rely on simple regression analysis to determine which predictor variables have a significant effect on ADHD, reporting interpretable coefficients and confidence intervals (CI) based on large sample MCS data [32]. These insights are important for validating biomarkers yet prediction performance is limited considering the potential range of approaches for processing, modelling and evaluating these *big data* [47], [52]. Machine learning explores and optimises these approaches, allowing more complex analysis of feature interaction and non-linear relationships, especially suited for large multi-variate datasets [47], [71], as seen in the wider literature [12], [13].

Predicting ADHD with machine learning describes a classification task where supervised learning trains models on predictor variables and class labels, typically clinical diagnosis [13]. Impressive performance figures have been reported in the field [13], [20], yet establishing a valid, reproducible and ethical machine learning framework, or pipeline, for testing temporal and duration effects on performance requires correct data processing, model selection and evaluation [71]. However, the literature reveals challenges with sample size, confounders and bias threatening validity, while interpretability and equality – important factors for real-world deployment; component to the TRIPOD+AI framework [5], [19] – are often overlooked [12], [13], [20].

Interpretable and equitable modelling

Interpretable machine learning provides transparent, auditable decision making and indicate implicitly how each input variable – biomarkers, ethnicities etc. – contributes to the output's variance [48], crucial for ethical representation and accountable clinical practice [5]. Basic et al critique weak interpretability in the field for undermining biomarker validity and black-box models prone to algorithmic bias [12]. However, less-interpretable models, e.g. ensemble/non-linear Support Vector Machines (SVM)/neural architectures, tend to capture greater complexity and superior performance [37], [72], [73], presenting a trade-off [48]. As recognised by the NHS funding trials in deep-learning for image-based diagnosis [16], although regulatory bodies

(NICE, British Medical Association) [15], [17], and UK legislation (*General Data Protection Regulation*, GDPR) [18], increasingly seek interpretable, or at least explainable, solutions [74], such as regression, linear-SVM and tree-based models seen regularly in the literature [12], [13].

Algorithmic bias defines systematic errors due to biases in data, labels or modelling [75], even interpretable approaches are susceptible where healthy performance metrics on the whole dataset may belie performance discriminations between subgroups, potentially compounding existing inequalities [76]. Data biases are somewhat mitigated with appropriate, large-cohort sampling. Label biases, e.g. due to disproportionate misdiagnosis in ethnic minorities [5], and biased modelling are harder to pre-empt and are seldom considered in the literature, although interpretability can help with the latter [12], [13]. A post-training solution is presented by Liou et al who perform calibration analysis on clinically deployed models identifying performance disparities (misdiagnosis) across sex and race, they also show how recalibration methods can reduce this [38].

In summary, transition to clinical practice requires transparent and well-documented workflows built on samples truly representative of the population [12], [13], [20], [21]. While less interpretable healthcare models are being explored, the focus remains on interpretability as demonstrated with the only currently accepted model used for facilitating ADHD diagnosis, QBtest, based on logistic regression of attention related tasks and optical movement sensing for activity levels [77]. Methodology for clinically useful ADHD prediction therefore depends on constructing valid biomarkers, equitable data modelling and ensuring repeatable results with a transparent, reproducible machine learning pipeline for testing the RQs.

3. Methodology

3.1 Philosophical approach

Interpretivist approaches have revealed the qualitative experience of those living with delays to care; Holden and Kobayashi-Wood describe how participants grieve for the lives they could have led if diagnosed sooner [78], while Gudka et al document the individual and systemic stress resulting from primary care shortcomings [79]. In contrast, prediction research aims to improve people's experience by expediting diagnosis through quantitative methods and sits firmly in the post-positivist paradigm conducive to generalisable, repeatable and objective results required for clinical application [80], [81].

This study therefore adopts an approach sympathetic to the lived-experiences of those with the condition and the ethical standards required for real-world use. The post-positivist philosophy enables an empirical and externally valid study for constructing biomarkers [12], and predicting ADHD while being guided by complementary interpretivist research towards ethical methods including non-invasive data collection sources, interpretable modelling and equitable processing in light of known subgroup inequalities. Furthermore, the limitations of post-positivism are recognised, for example selecting a clinically acceptable modelling threshold for misdiagnoses (false negatives/positives) is a subjective decision to be considered by medical professionals through a constructivist discourse [82]. To aid such decisions, however, a range of thresholds is presented and discussed.

3.2 Design

A quantitative approach is selected to process secondary observational data for an ADHD predictive modelling study. A supervised machine learning pipeline is designed to analyse performance across different feature, temporal and subgroup conditions. The unit of analysis are the participants, represented by objectively measured activity-derived variables, ADHD covariates and a binary class label based on clinical diagnosis operationalised from parent-questionnaire responses. The activity variables are derived from accelerometer data, daily aggregates of accelerometer counts handcrafted to reflect activity known to differentiate ADHD, and automatically extracted using the tsfresh library which produces a large number of generic statistical features based on the raw data. Handcrafted are simple and easily interpreted based on theory, whereas tsfresh can explore intra-daily patterns and greater complexity. Feature importance comparison helps to answer RQ1 [83].

For testing RQ2, the pipeline builds classifiers based on stratified/non-stratified conditions for both handcrafted and tsfresh sets, with feature-selected sets for each condition compared with non-parametric tests. These conditions were also tested across a range of learning algorithms to assess model performance and interpretability, with coefficient analysis. Inclusion of covariates with the selected features demonstrated whether activity features can add to clinical assessment, and subgroup analysis assessed for discrimination and miscalibration. The pipeline then served to compare different monitoring period durations, answering RQ3, in order to make novel recommendations for data collection, especially relevant in school-age children known for significant attrition and non-wear rates [26], [95].

The quantitative approach allows for a standardised sample, pipeline and evaluation scheme to be established for making repeatable, like-for-like comparisons across the various research conditions. The large-cohort sample and objective, numerical features are suitable for statistical significance testing and generalising to the UK population. However, the observational nature of the study limits causal inference [84], relevant for determining covariate relationships with ADHD, such as correlated risk of higher male prevalence [8], [24].

3.3 Methods and Computational Approach

3.3.1 Data Source and Environment

Secondary data is sourced from the MCS via the UK Data Service [26]. The MCS conducts longitudinal research on 18,827 children born around the millennium, collecting sweeps of observations, questionnaire and interview data every 2-3 years. The large-cohort includes stratified samples of electoral wards, over-sampled to increase minority representation for ethnicities, regions and socio-economic groups [26], facilitating equitable processing and generalisable results [32]. Questionnaire responses include sex and health data used for controlling confounders and extracting ADHD related variables as class labels for supervised learning [48]. Natural-setting activity data was collected for 7 days at age-7 with a waist-worn, non-invasive GT1M actigraph [85], with n=6675 valid responses for deriving ecological biomarkers [55].

Computation environment

Data was processed in a secure Python v3.11 environment with NumPy and pandas for data management, scikit-learn for machine learning tasks, statsmodels and SciPy for additional statistical tests, and Matplotlib and seaborn for visualisation [86], [87], [88], [89], [90], [91], [92], [93]. Computation conducted on a local device with 40GB memory and 20 logical cores, oversize datasets were chunked and heavy compute workloads parallelised where possible [94].

3.3.2 Cohort Sampling and Preprocessing

Machine learning algorithms typically require complete, cleaned data [71]. Participants missing reliable actigraphy were excluded [27], as were those missing covariate data as confounders could not be ruled out. Features with modest missing values were subject to preprocessing rules for imputation executed at training time to ensure imputation occurred within-fold, keeping cross-validation scores free from leakage [48]. Categorical features imputed the most frequent value and were ordinal encoded, numerical features used medians, standard practice and robust to outliers [47], [71]. An extreme value threshold has also been statistically derived through quality control research [95], implicitly applied to the MCS derived daily aggregates

used for handcrafted features and included explicitly in this study's raw timeseries processing. Final datasets were then standardised within-fold to even-weight features for optimal learning, as Witten et al recommend [47].

3.3.3 Feature Engineering and Selection (RQ1/RQ2)

Both daily aggregate data and raw timeseries data were explored for deriving ADHD biomarkers, focusing on the most promising phenotypes presented in the literature, such as time in sedentary activity, inter- and intra-daily variances and circadian rhythms [14], [32], [54]. Features handcrafted from the MCS provided aggregates for the four activity intensity groups, based on the actigraph's basic unit of activity, counts-per-minute (CPM) [96]: sedentary, light, moderate and vigorous. Total counts were also included and all features were standardised against total wear-time for the day to give relative, comparable values. The standard deviation was calculated across days for relative total counts to measure inter-daily variance, a standard measure in this field [12], [14], and a variable for the month data was derived to aid seasonal modelling. Daily aggregates however cannot reveal intra-daily patterns such as school/home differences known to differentiate ADHD [25], [41], [42], [54].

The raw timeseries data provides counts every 15 seconds and therefore appropriate for far more granular analysis of within-day features, appropriate for testing RQ1. However, in order to process this data it needed to be mapped to the valid and reliable sample first, this required development of two algorithms. The first applied a non-wear marker to each participant's timeseries data, based on the same rules used by MCS to remain consistent [27] as non-wear algorithms have been shown to affect sedentary time estimates [97]. The second determined how many days offset the timeseries' were from the first recorded valid day, a marker could then be applied against the raw data for all valid days, plus a weekend marker necessary for stratifying weekday/weekend features. Automatic feature extraction library, tsfresh was then applied to the sample, producing 778 statistical features for each valid day [58]. Applying across longer periods was unsuitable given valid days were not always contiguous.

For testing RQ2, per-day features from both the handcrafted and automated set were then averaged across either all days or for the weekday/weekend split, such that both approaches can compare temporal stratification effects against the non-stratified equivalent, tested with

Wilcoxon signed-rank (WSR) on paired cross-validated AUROC differences, similar to others [37], [41]. Other methods for processing raw actigraphy timeseries data in the literature include the use of deep learning [98], [99], which lie beyond this study's scope for interpretable methods.

The large generic output of tsfresh is then subjected to a series of feature reduction methods to mitigate known problems for high-dimensional datasets [47]. A first pass removes features with constant values and applies a correlation filter for removing redundant, collinear features [100]. Features are then selected for consistently ranking well across a range of methods based on mutual information, f-scoring, regularised logistic regressors and recursive feature elimination [83], [101]. Methods dependent on a base classifier were configured with both a regularised logistic regressor and an extra trees classifier, both of which performed well during exploratory data analysis (EDA) and cover interpretable and non-linear feature spaces respectively [48]. This range prevents any particular method dominating and applies broad assessment of feature importance. Handcrafted features were likewise subjected to feature selection methods, albeit starting from an already modest set.

The selected features were then combined with covariates, including parent responses to the Strengths and Difficulties Questionnaire (SDQ), and class labels.

3.3.4 Modelling and Evaluation

Classifiers often reported in ADHD prediction research [12], [13], [14], and found to perform well during EDA, were chosen for model comparison: regularised Logistic Regression and Naïve Bayes offer interpretable features and ensemble-based Extra Trees, while less interpretable can learn non-linear relationships and feature interactions [48]. SVM, Random Forest and XGBoost were also initially included but exploratory testing found mediocre results unable to justify excessive training times. A base-line model predicting majority class was also included [102]. After model selection, hyperparameters were tuned with the grid search method, avoiding additional overhead from more complex methods such as Bayesian [103].

Sub-group analyses were then considered to examine potential bias based on covariates associated with ADHD inequalities discussed in the literature review. Unfortunately despite MCS oversampling, the positive class size ($n=6$) for ethnic subgroups was too small for reliable analysis [104], [105]. The sex subgroup, however, was found to be more evenly distributed and

thus appropriate for sensitivity analysis [106]; out-of-fold probabilities for each sex were estimated by the final selected model and assessed across a range of fairness metrics – Brier scores, calibration curves and sensitivity/specificity thresholds – as shown to support clinical decision-making in diverse populations [13], [19], [38].

The low positive class in general also precludes hold-out testing, robust to overfitting [48], due to potential sample bias [107]. Instead, final metrics are reported on 5-fold class-stratified cross-validation to retain a proportional and valid class distribution [37] and common practice in the field for balancing bias/variance trade-off [12], [48]. Moreover, to compensate for lack of hold-out testing cross-validation for feature selection and final evaluation was resampled and repeated 5 times, improving precision and generating a measure of the model's error variance, important for demonstrating model consistency and reproducibility of the results.

Modelling is primarily evaluated on AUROC, standard in and therefore comparative to other ADHD prediction studies [12], [13], robust to class imbalance [108], and preferred over threshold-based metrics for not assuming a clinical threshold [109], consistent with this study's philosophy. AUROC indicates how well ranked each class is overall, supporting metrics Area Under the Precision/Recall Curve (AUPRC) [110], confusion matrix and sensitivity/specificity thresholds aid further analysis, as is standard [12], [13].

3.3.5 Monitoring Period Investigation (RQ3)

To compare monitoring period durations for RQ3, participants with at least six valid days monitoring were selected, seven or more saw an unacceptable drop off in sample size, see figure 3.1. Feature sets were then prepared using two to six days data for each participant, maintaining a minimum and typical distribution of weekdays and weekend days. The finalised pipeline – preprocessing, selected features and tuned model based on the full data sample – was then applied using identical cross-validation folds for each duration. This allowed differences in AUROC and AUPRC between each reduced duration and the 6-day reference condition to be assessed using paired bootstrap resampling, appropriate for non-independent samples [111]. The bootstrap process resamples the data 5,000 times in order to populate a distribution of results for each duration condition, generating confidence intervals and data to calculate Holm adjusted p-values for statistical significance testing [112], [113].

Similar AUROC values may mask instability for individual predictions, therefore individual stability between durations was assessed with the widely used Spearman rank [114]. Thus, duration effects were evaluated both at the group-performance level and at the individual prediction-stability level, responding to the research question.

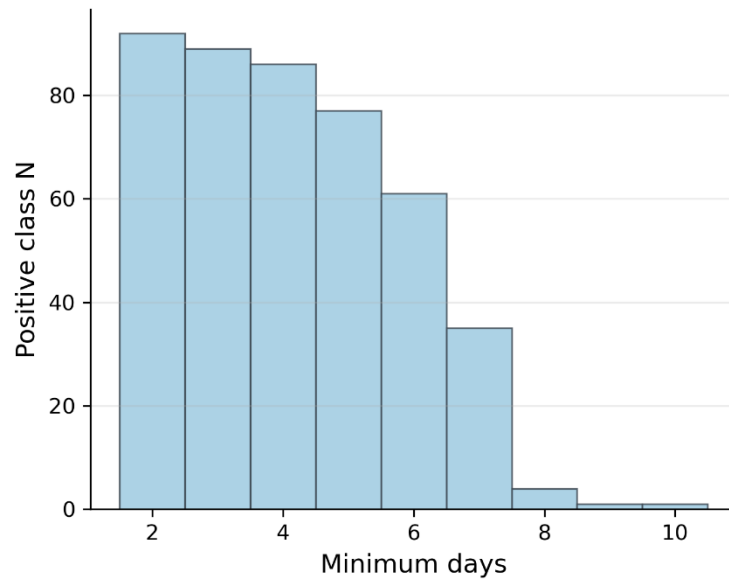


Figure 3.1. Sampling based on minimum valid days per participant. Six days retains over 60 positive class labels, whereas seven or more days considered less reliable at < 40

3.4 Statement of Ethics

Ethical approval for this project was granted by the University of York's ethics committee prior to data access. The primary data source documents the safeguards protecting participant welfare in their ethics review [115], confirming participant consent and non-invasive data collection methods in line with University and Professional codes of ethics for the avoidance of harm [116], [117], [118].

The terms and conditions from the data provider, UK Data Service, have been adhered to and are compliant with the relevant legislation, UK GDPR [119]: risk of reidentification is considered to be 'sufficiently remote' by the data provider with data already anonymised [120], essential data for the project has been stored and transmitted securely following the University Information Security Policy [121], and UK Data Service's user agreement [120]. Data is used solely for the intended purpose of academic research where all results are reported as aggregates or statistics to further protect individual exposure, datasets are set to be destroyed on project completion [120].

The methods of the study follow the TRIPOD+AI framework for transparent and accountable data processing and reporting, where interpretable machine learning and subgroup analysis support integrity and validity of the results, ensuring British Computer Society (BCS) core principles for equality and making information technology for everyone are upheld. The motivation for the project aligns with Association for Computing Machinery (ACM) ethical principles for making a contribution to society [118], looking to see how state-of-the-art technologies can be applied to help those waiting for healthcare and support an overburdened public healthcare system.

4. Results

Descriptive results of processing and sampling the data for the various stages of the study are presented in table 4.1, below. The final sample for feature, model selection, subgroup, covariate and threshold analysis includes 92 ADHD + 4,941 controls, exceeding the minimum sample size suggested by Pulini et al, even if equally subsampled [20]. This is based on the most constrained sample requiring at least one weekday and one weekend day in order to stratify these groups, applied to the non-stratified condition also for like-for-like comparison. Consistent with the literature, males are three times more prevalent than females in the sample [24], the low non-white prevalence for the positive class precludes meaningful analysis for ethnicity subgroup. The 6-day constraint reduces the monitoring period sample further yet remains sufficient with positive class n=61.

	Class prevalence		Total (N)
	Control	ADHD	
Valid actigraphy responses	6,525	150	6,675
Excluding missing values	6,649	150	6,649
Excluding confounders	6,183	111	6,294
Stratified sample	4,941	92	5,033
Subgroups			
Sex:			
Female	2,473	22	2,495
Male	2,339	66	2,405
Unknown	129	4	133
Ethnicity:			
White	4,339	86	4,425
Non-white	602	6	608
Monitoring period analysis	3,670	61	3,731
	Days	Data points (epochs)	
Median actigraph duration	6	34,560	
Total data points processed		171,598,928	

Table 4.1 Descriptive statistics on participant and actigraphy samples

4.1 Features (RQ1/RQ2)

Following feature selection across the four main conditions, table 4.2. reveals stratifying on weekday/weekend for the tsfresh set improves AUROC substantially ($p < 0.0001$) over the non-stratified set and the handcrafted sets, with best result Logistic Regression 0.815 ± 0.041 . A very mild improvement is seen for the handcrafted stratified set over its non-stratified equivalent with Naïve Bayes but the difference too small and variance too high to be reliable. The stratified-tsfresh set was therefore selected for final pipeline, downstream analysis – the large set of features unable to be reduced further without loss of performance. The covariates alone make modest gains over the dummy classifier and marginally support the stratified-tsfresh set with AUROC gains of 0.006 and tighter variance. Addition of SDQ improves performance further to 0.878 with greater stability/reduced variance. AUPRC was consistent with AUROC scoring although noisier as is more sensitive to the low positive class prevalence [110], see appendix tables B1 and B2 for full results.

Condition	Features		AUROC			
	initial	selected	Logistic Regression	Naive Bayes	Extra Trees	Dummy
Handcrafted						
non-stratified	8	5	0.652 ± 0.064	0.666 ± 0.066	0.606 ± 0.057	0.5
stratified	13	5	0.649 ± 0.065	0.672 ± 0.060	0.540 ± 0.051	0.5
tsfresh						
non-stratified	778	273	0.722 ± 0.056	0.652 ± 0.062	0.617 ± 0.058	0.5
stratified	1554	442	0.815 ± 0.041	0.644 ± 0.066	0.604 ± 0.072	0.5
difference (p)			0.093 (< 0.0001)			
Covariates						
Sex, Ethnicity, BMI	3	-	0.606 ± 0.063	0.593 ± 0.066	0.570 ± 0.052	0.5
+ tsfresh-stratified	445	-	0.819 ± 0.039	0.643 ± 0.063	0.604 ± 0.052	0.5
+ SDQ	446	-	0.878 ± 0.035	0.737 ± 0.064	0.659 ± 0.065	0.5

Table 4.2. Exploratory results for handcrafted vs tsfresh extracted features, weekday/weekend stratification and covariate analysis

Logistic Regression coefficients for the covariate analysis show Sex most prominent ($\beta=0.504$), Ethnicity less so ($\beta=0.267$) while BMI (Body Mass Index) contributed relatively little signal ($\beta=-0.019$), see appendix table B2. When covariates combined with tsfresh-stratified, only Sex remained important ranking 30/445 features. Incidentally, the feature representing month data collected was not selected throughout the feature selection process and features selected for the handcrafted set were not consistent across selection methods (appendix table B1).

Interpreting the tsfresh features is less straight-forward; they are generic and many. However, looking at feature-families is insightful: a balanced split among weekday ($n=222$) and weekend ($n=220$) is found; and table 4.3 shows a descriptive grouping and count of the selected features based on feature names and tsfresh documentation [122], see appendix table B3 for details. The most selected features by far are frequency and spectral-based such as Fourier derivations which reflect periods, oscillations and rhythms, with smaller contributions from distributional, variability, timing, temporal-dependence, and complexity features. Interestingly, the least prominent selected include the simpler sum/magnitude features most closely related to the handcrafted features.

Feature family	Selected count (% retained)	% coefficient contribution	Example features	Interpretation
Frequency / spectral structure	345 (38%)	83.8%	fft_coefficient, fourier_entropy	Rhythmic, periodic or frequency-domain structure in activity counts
Change, variability, and sequence structure	28 (13%)	4.6%	change_quantiles, longest_strike_below_mean	Local variability, repeated values, and sustained activity/inactivity patterns
Distribution, extremes, and bursts	15 (17%)	3.9%	large_standard_deviation, skewness, kurtosis	Spread, tails, unusual values, and short bursts or peaks of activity
Timing, trend, and energy location	19 (13%)	3.8%	linear_trend, first_location_of_maximum	When activity occurs within the monitoring window and whether activity changes over time
Temporal dependence and stationarity	13 (18%)	2.4%	autocorrelation, augmented_dickey_fuller	Persistence, regularity, lagged dependence, and stability of the time series
Complexity / entropy	5 (19%)	0.6%	lempel_ziv_complexity, permutation_entropy	Irregularity, complexity, and unpredictability of activity patterns
Other / miscellaneous	1 (25%)	0.5%	benford_correlation, query_similarity_count	Features that do not fit cleanly into the main methodological families
Simple activity volume / magnitude	7 (18%)	0.4%	sum_values, abs_energy, mean, median, maximum	Sum amount, level, or magnitude of activity counts
Nonlinear dynamics / model-based	0 (0%)	0.0%	friedrich_coefficients, max_langevin_fixed_point	Nonlinear model-based descriptors of time-series dynamics

Table 4.3. Selected tsfresh features grouped into families for interpretability

4.3 Model Selection

Table 4.2 also reports the performance across classifiers. Logistic Regression consistently outperformed Naïve Bayes and Extra Trees in all conditions with the selected tsfresh-stratified features and across all metrics. Logistic Regression was therefore selected for the final pipeline, hyperparameters were optimised to include an L2 penalty and $c=0.1$ for regularisation, appendix table B4. Final scoring achieved AUROC 0.846 and AUPRC 5.9x baseline (baseline is equivalent to a dummy classifier), see figure 4.1.

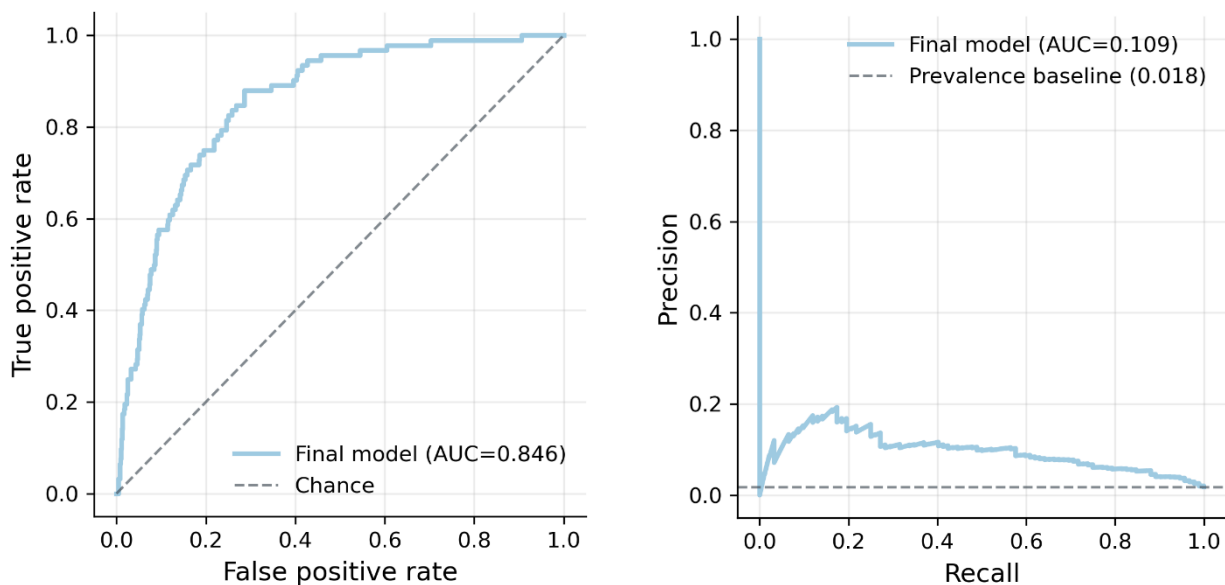


Figure 4.1. Final model outputs, left: Receiver Operating Curve (ROC) plot, right: Precision-Recall (PR) curve – note the low precision scores reflecting high false positives, information that is absent from ROC analysis alone

True-positive rate (sensitivity or recall) and false-positive rate ($1 - \text{specificity}$) are threshold dependent – a choice can be made to either optimise sensitivity and avoid false negatives – useful for screening – or optimise specificity and avoid false positives – good for confirming diagnosis [123]. Threshold analysis, appendix table B5, reveals a balanced threshold value of 0.01 achieving sensitivity 0.837 and specificity 0.739, meaning 83% of true positives are selected but 26% of true negatives (controls) are also incorrectly reported as ADHD, significant given the greater proportion of controls. As higher specificities are selected, e.g. at threshold

0.023 giving specificity > 0.8 , sensitivity declines to 0.739. The ROC plot, figure 4.1 (left) presents the trade-off across all thresholds, where the PR curve (right) shows how the proportion of true to false positives (precision) changes across sensitivity (recall), useful for imbalanced data as a well-balanced sensitivity/specificity may conceal the large proportion of the majority class, e.g. at sensitivity 0.837 as above, precision is < 0.1 indicating over 10x controls incorrectly classified for each correct positive. Note, threshold results are exploratory and may be optimistic as have been calculated outside of the pipeline's cross-validation process.

The final model is demonstrated in figure 4.2, note the concentration of positive class toward the top of the model illustrating class separation with positive class probabilities typically ranked higher. The selected threshold suggests a cut-off that could be used to discriminate class with accepted misclassification on either side of the line.

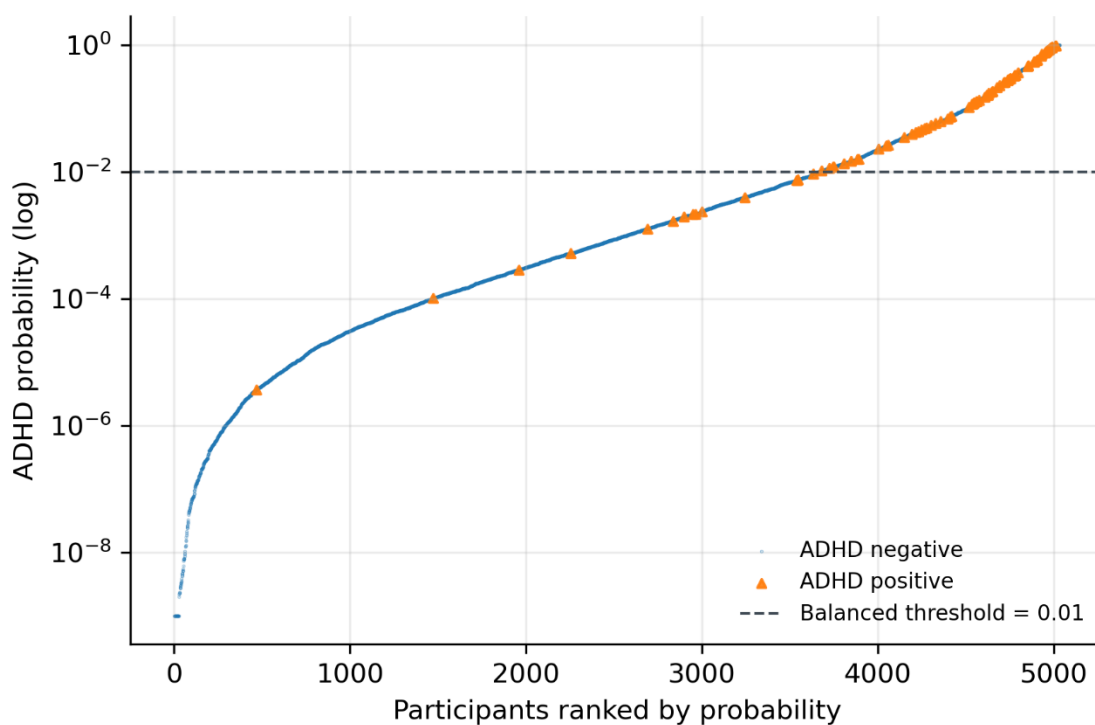


Figure 4.2. Final model class probability rankings

4.4 Subgroup: Sex

Table 4.4 shows female AUROC 0.881 higher than male 0.799 and AUPRC 5.5x baseline compared with males 3.9x. Female discrimination appears more specific than males too with better sensitivity and specificity at this threshold (0.01, selected to balance the All group, see appendix table B6), low precision scores represent many false positives across all groups. All calibration intercepts are negative and slopes substantially below 1 in both groups, see figure 4.3. Lower Brier score seen in females.

Group	Prevalence	AUROC	AUPRC	Brier	Calibration	
					intercept	slope
All	1.8%	0.834	0.082	0.035	-2.534	0.299
Female	0.9%	0.881	0.048	0.023	-3.019	0.319
Male	2.7%	0.799	0.107	0.049	-2.320	0.275

Group	Sensitivity	Specificity	Precision	TP	FP	TN	FN
All	0.761	0.750	0.053	67	1,201	3,611	21
Female	0.818	0.799	0.035	18	496	1,977	4
Male	0.742	0.699	0.065	49	705	1,634	17

Table 4.4. Fairness metrics for sex subgroup analysis, lower table at threshold 0.01. TP=True Positive, FP=False Positive, TN=True Negative, FN=False Negative

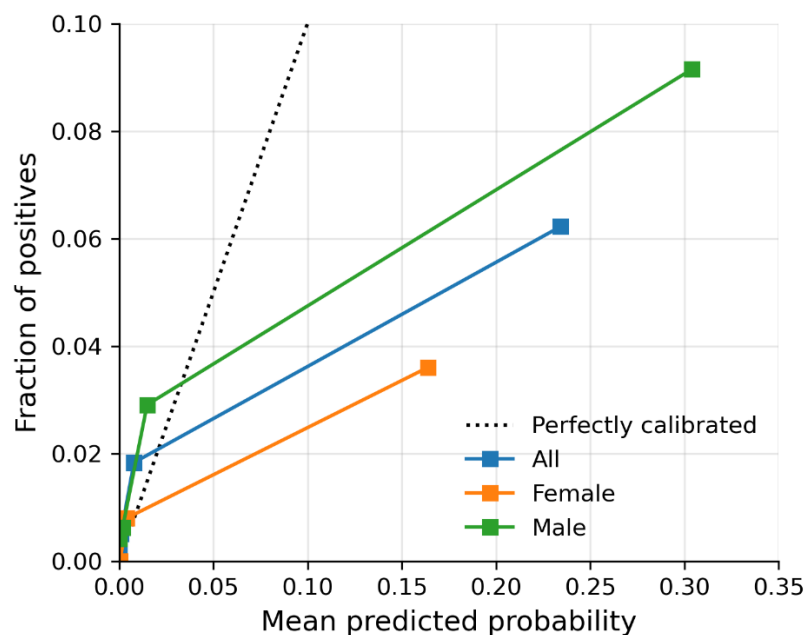


Figure 4.3. Calibration plot for sex subgroup analysis, note x-axis scaling

4.5 Monitoring Period Results (RQ3)

Performance improved monotonically across AUROC, AUPRC and Spearman metrics as duration increased, table 4.5. Figure 4.4 illustrates 2- and 3-day durations both substantially lower than the 6-day reference, with upper-bound CIs wholly below the 6-day lower-bound, and the difference found to be significant with p-values < 0.001. 4 days showed intermediate performance, with a modest reduction in AUROC and high rank agreement (0.74) with the 6-day predictions. 5 days produced near-equivalent performance and found to not be significantly different with p-value 0.688 and Spearman agreement 0.863. AUPRC showed the same general trend but with wider uncertainty, consistent with the instability of precision-recall estimates under low outcome prevalence.

Days	AUROC				AUPRC				Spearman
	AUROC	CI	Δ vs 6	p-value	AUPRC	CI	Δ vs 6	p-value	
2	0.598	0.522 - 0.676	-0.186	< 0.001	0.023	0.016 - 0.034	-0.04	< 0.001	0.418
3	0.624	0.553 - 0.693	-0.160	< 0.001	0.037	0.021 - 0.084	-0.03	0.149	0.542
4	0.730	0.679 - 0.780	-0.054	0.070	0.054	0.031 - 0.110	-0.01	1	0.744
5	0.778	0.744 - 0.811	-0.007	0.688	0.057	0.037 - 0.106	-0.01	1	0.863
6	0.785	0.726 - 0.841	-	-	0.063	0.041 - 0.114	-	-	1

Table 4.5. Monitoring period duration comparison

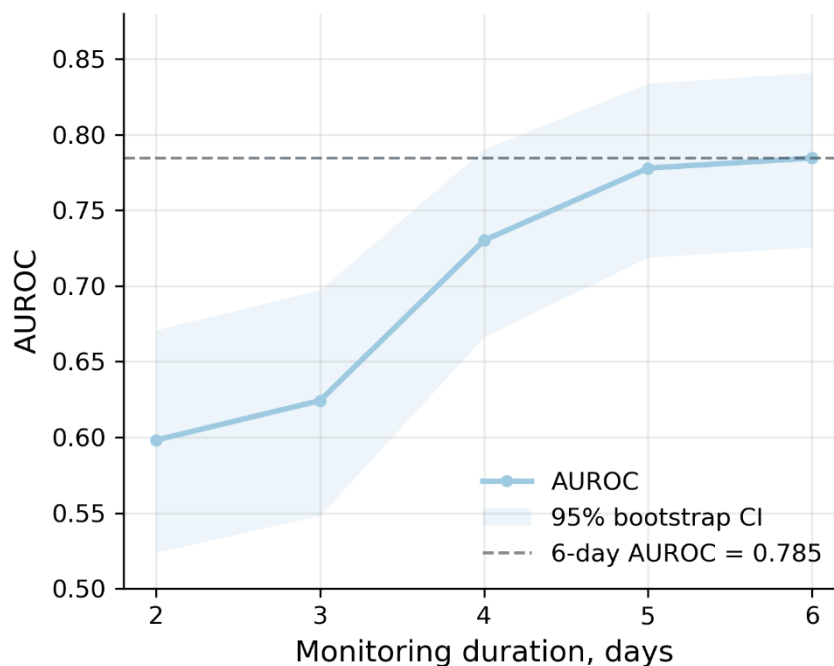


Figure 4.4. AUROC by monitoring duration with CI

5. Discussion

Feature Interpretation and Temporal Effects (RQ1/RQ2)

The handcrafted features demonstrate a modest ability to predict ADHD (AUROC 0.672) on five selected features but are substantially outperformed across all metrics and models by the automatic-extracted feature sets (AUROC 0.815), indicating that aggregate means and variances alone miss key information. Analysing tsfresh feature importance, table 4.3, reveals features similar to the handcrafted in *simple activity volume and magnitude* family contribute very little compared with the more granular and complex frequency/spectral features, periods and non-linear distributional metrics such as kurtosis and skewness, perhaps explaining the limitations of the handcrafted set, consistent with the significant yet low effect-size features reported by Brandt et al [32].

The results demonstrate the importance of intra-daily patterns in ADHD activity, providing a descriptive response for RQ1, and present an interpretable alternative to neural approaches seen in the literature [98], [99]. Mapping the features back to ADHD phenotypes is not exact, although the many low-level Fourier coefficients selected have been shown to encode low-level temporal frequency structure [124], suggesting the signal is from activity and not from, for example, features related to device non-compliance (e.g. extremes, irregularity). Being able to interpret features therefore aids validating biomarkers and provides accountability that correct phenotypes are being assessed, important for building the confidence necessary for clinical adoption [5], [21].

Statistical testing ($p < 0.0001$) and the substantial effect size (AUROC +0.093) of stratifying feature-selected tsfresh features demonstrates the significance of weekday/weekend differences in ADHD prediction, answering RQ2. The results are further supported with a near-equal selection between weekend/weekday features, suggesting signal specific to each strata adds predictive power, supporting work by Pedersen et al and others on the significance of weekly structures on ADHD presentation and in particular how these effects play out throughout the day [25], [41], [42], explaining perhaps why these differences were not pronounced between the daily-aggregated handcrafted sets. Conversely, the significance of seasonal structures found in the literature [45], [46], cannot be verified at this time as relevant data for term-times is not available in MCS and the approximation using month data collected found to

be insufficient to make an impact on results. The positive results from intra-daily and weekend features have proved significant, demonstrating the benefit of working with raw actigraphy timeseries data and not relying solely on daily aggregates [32], [37], and the advantage of collecting data across both weekdays and weekends.

Model Interpretability and Performance

The most interpretable model in this study, Logistic Regression, also outperformed less interpretable ensemble models, table 4.2, indicating gains from non-linear patterns and feature interactions were unrealisable. Interpretable Logistic Regression coefficients helped to rank the feature-families, table 4.3 (appendix table B3), and facilitate covariate analysis, below. Final, tuned model scores of AUROC 0.846 and AUPRC score 5.9x above baseline demonstrate strong discrimination between classes, supported visually in figure 4.2, and align with realistic expectations set by other reliable studies [20], [32], [37], validating this study's pipeline for subsequent subgroup and monitoring period analysis.

Analysing ROC and PR curves over various thresholds, figure 4.1, shows how the model could be useful in the real world. Optimising for sensitivity could, for example, help justify a screening solution; those on long waiting lists for assessment could be screened with at-home actigraphy monitoring and those classified at risk sent a care bundle with information and support pre-diagnosis, mitigating unsupported consequences [5]. Precision scores at the chosen threshold reveal the risk of controls being misclassified to be high but the cost of sending a bundle is low – selecting a threshold can find an optimal cost/benefit for this solution. This also offers an alternative or supplement to self-screening, known to over-report [4]. Conversely, the cost of misdiagnosing and starting treatment for those who do not really have ADHD is high, here specificity can be optimised to reduce positive-class misclassification.

Covariate and Subgroup Analyses

The relative coefficients for Sex and Ethnicity in the covariate only model and their modest contribution to the +tsfresh-stratified model, table 4.2, support their association with ADHD as found in the literature [5], [8]. In contrast BMI was not found to contribute meaningful signal suggesting findings by Brandt et al are limited to the small effect size they report [32]. The findings support a model for deployment that incorporates Activity, Sex, Ethnicity and SDQ – all of which contribute to improved ranking and stability – although further causal analysis is recommended to better understand potential algorithmic bias [75].

Higher AUROC and relative AUPRC score for females, table 4.4, suggest moderately stronger rank-based discrimination compared to males. The confusion matrix at this threshold reveals the extent of the low female positive class prevalence which may lead to instability in these results – for example, sensitivity would drop below the male score if just 2 more true positives were misclassified. Likewise, lower female Brier scoring typically indicates better calibration but this is potentially too affected by the lower prevalence to be conclusive [125]. Negative calibration intercepts across all groups indicate systemic overprediction as seen in the high number of false positives, with stronger overprediction in females by a small factor. Calibration slopes were substantially below 1 in all cases, figure 4.3, revealing the scale of miscalibration of probabilities, slightly worse in males.

The analysis reveals broadly comparable subgroup performance, with imbalance seen in higher female ranking metrics but also unstable due to prevalence. The model can therefore rank well across males and females, fairness could be mildly improved between males and females although not in the direction expected given higher risk prediction in males. The model's probabilities are substantially under-scaled and thus do not represent actual risk. Future work could see mitigation strategies correct the miscalibration, bridge the small gender gap and realise more informative individual probability predictions to aid clinical decision making [38].

Monitoring Period Investigation (RQ3)

The study finds longer duration improves prediction rankings (AUROC, AUPRC) and predicted probability stability (Spearman), figure 4.4 demonstrates when these gains reach diminishing returns and when they are significant enough to materially affect prediction performance, responding with statistical significance testing to RQ3. Similar trends have been reported on physical activity reliability by Rich et al, also on MCS data [27], yet results specific to ADHD for duration and reliability are novel here, table 4.5. These results seek to address a gap in ADHD prediction research; many different durations have been studied but not in a controlled and comparable way and seldom in the context of the ethical considerations discussed above. The results inform that daytime actigraphy can produce reliable predictions with as few as 5 days monitoring, this may be useful when longer wear times are challenging due to stigma, physical sensitivities or non-compliance [26], [55], [126]. The significantly reduced predictive power reported at 2 and 3 days can advise against accepting data of this length under these conditions as the risk of misclassification significantly increases, rendering any application of the robust modelling approach presented here unreliable whether for screening, diagnosis or further ADHD prediction research.

Threats to Validity

Label bias

Labels from the MCS data reflect a time when prevalence was lower (2.2%, table 4.1) due to stricter criteria and underdiagnosis, relative to the standards today (c. 5%) [3], [4]. Therefore at least as many controls as there are positive labels are likely to be mislabelled, confusing learning. Future work ideally operationalises labels closer to ground truth.

Sample bias

ADHD and Sex distributions remained consistent across sampling, table 4.1, further analysis of how sampling may have affected other demographics, known to deviate in such studies [32], [127], should be considered in future work.

Construct validity

Representing a heterogenous condition with 6-day, daytime-only activity biomarkers from 7-year olds is a limitation of the primary data, prior research has established the importance of persistence of symptoms over time [7], night-time patterns [37], [43], non-motion phenotypes, and age to be significant for discriminating ADHD and capturing the range of symptoms across subtypes/presentations [2], [24]. Constructing multi-variate datasets inclusive of these features will help improve alignment with assessment criteria and prediction performance [12], [13], [21].

6. Conclusion

This study has investigated the effects of actigraphy duration and temporal factors on ADHD prediction with data generalisable to UK healthcare, aiming to provide insights into optimal data processing and demonstrate how interpretable and equitable machine learning can be applied to aid the under-diagnosis crisis.

Main findings

MCS activity data is able to differentiate ADHD from controls using Logistic Regression tuned to AUROC 0.846, on biomarkers extracted using automatic feature extraction techniques. These techniques were found to be superior to handcrafted daily aggregates, suggesting the granular intra-daily features available from the raw actigraphy timeseries, in particular the frequency and spectral-based features, are important for differentiating ADHD (RQ1). Interpretable methods are important for helping to validate these biomarkers and rule out bias from non-activity artefacts such as monitoring device non-compliance. Model interpretability further aids analysis of feature coefficients, including impact of ADHD covariates

Subgroup analysis finds the final model ranks well for both males and females but with marginal discrimination favouring females, although concludes the margin unstable due to low class and gender prevalences. Individual predicted probabilities found to be substantially miscalibrated for all sexes whether separately or combined. Weekend stratification proved significant to performance gains (RQ2) supporting research that ADHD physical activity symptoms are dependent on weekly temporal structure. No clear evidence was found for monthly or seasonal structures although this is possibly due to unavailable data. Longer monitoring period durations significantly improve model performance ranking and stability (RQ3), with good performance found with a minimum of 5 days and small gains up to 6 days. 2 and 3 days were significantly worse than the 6-day reference, 4 days showed compromised performance.

Recommendations

These findings promote large sample, interpretable methods for valid processing and modelling of activity data for predicting ADHD, and to consider calibration/sensitivity methods for

assessing subgroup results to flag algorithmic bias/inequalities, especially to help translate prediction technologies into an accountable, clinical environment rightly sceptical of overly optimistic scorecards. This study also recommends working with raw actigraphy data and using tsfresh for extracting high dimensional timeseries features for capturing the complex, intra-daily biomarkers aggregate data may miss. Additional temporal structure is found in the difference between weekdays and weekends, capturing and conditioning on this data proved an effective method for significantly improving prediction scores. In addition, the study recommends at least 5 days actigraphy for reliable predictions, less than 4 days significantly worsens prediction stability and ranking.

These methods and results provide novel insights into temporal effects in ADHD prediction research, bridging a research gap by realising these findings on a diverse large UK cohort with transparent methodology. The author hopes to facilitate a technology-driven solution to support those waiting for an ADHD diagnosis, expedite assessment and prevent misdiagnosis, thereby alleviating associated harms to individuals, their families, the healthcare system and wider society.

Limitations and future work

Despite the large MCS sample, ADHD's low prevalence constrained subgroup analysis. In addition, the MCS is UK-bound restricting generalisation beyond. Furthermore, the observational nature of the study limits causal analysis of covariates such as understanding whether Sex contributes to differences in activity patterns or simply predicts prevalence differences. This extends to other important factors such as subtypes, comorbidities and social-economic status, often excluded as confounders yet likely important parts of the causal structure needed to optimally model such a complex and heterogenous condition. Future work can consider these limitations and build on concepts unavailable to this study such as the benefits of continuous day and night monitoring with multi-modal fitness trackers as ecological, non-invasive biosensors for capturing a broader range of phenotypes. Longitudinal and highly-scalable data collection would help represent ADHD consistency over time and accommodate large, diverse populations. RADAR-base (King's College London), an open-source wearables data repository [70], presents an approach for addressing many of these needs and could be the basis for future work into taking ethical ADHD prediction research to the next stage.

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Appendix A: Artefacts

The agreed artefacts consist of the outputs from the project analysis, ethics approval documents, Python code for reproducing the outputs and the source data.

Artefact directory contents:

- Summary txt file of artefacts directory contents
- Ethics Approval submission documents and confirmation email
- Python notebook and YAML environment file
- Original data and precomputed feature datasets
- Excel document with code outputs for:
 - o Feature selection
 - o Model selection
 - o Model coefficients
 - o Hyperparameter tuning
 - o Final model predictions
 - o Sensitivity/specificity threshold analysis
 - o Subgroup analysis
 - o Monitoring period investigation

Appendix B: Supporting Material

Table B1. Feature selection full results from methods across handcrafted and tsfresh approaches, and for weekday/weekend stratified and non-stratified conditions:

Handcrafted - non-stratified set						
Starting features: 8						
selector	roc_auc_mean	roc_auc_std	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
mi_k5	0.648	0.064	0.042	0.016	0.567	0.659
f_classif_k5	0.648	0.064	0.046	0.021	0.567	0.657
elastic_selector	0.642	0.063	0.045	0.019	0.548	0.658
l1_selector	0.639	0.063	0.046	0.021	0.545	0.660
rfe_k5	0.635	0.062	0.048	0.025	0.541	0.659
none	0.633	0.062	0.045	0.022	0.530	0.660
f_classif_k10	0.633	0.062	0.045	0.022	0.530	0.660
Top three selectors' features						
mi_k5		f_classif_5		elastic_selector		
feature	score	feature	f_score	feature	coef	
MOD	0.004	CPM	36.198	CPM	0.270	
VIG	0.003	MOD	32.491	SED_SD	0.203	
SED	0.002	VIG	27.743	MOD	0.165	
CPM	0.002	SED	22.828	CPM_SD	-0.144	
LIT	0.001	CPM_SD	9.081	VIG	0.134	

Handcrafted - stratified set						
Starting features: 13						
selector	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
f_classif_k5	0.643	0.064	0.038	0.010	0.536	0.657
mi_k5	0.641	0.052	0.037	0.008	0.534	0.660
rfe_k5	0.626	0.062	0.038	0.012	0.504	0.657
f_classif_k10	0.625	0.057	0.039	0.017	0.509	0.660
l1_selector	0.618	0.055	0.040	0.014	0.489	0.660
none	0.617	0.054	0.040	0.017	0.490	0.664
elastic_selector	0.614	0.066	0.037	0.011	0.502	0.658
Top three selectors' features						
f_classif_5		mi_k5		rfe_k5		
feature	f_score	feature	score	feature		
CPM_WK	28.560	VIG_WK	0.006	MOD_WK		
MOD_WK	28.158	CPM_WE	0.003	SED_WE		
VIG_WK	22.288	LIT_WK	0.002	MOD_WE		
CPM_WE	20.576	MOD_WE	0.001	VIG_WE		
SED_WK	17.818	CPM_SD	0.000	CPM_WE		

tsfresh - non-stratified set

Starting features: 778

after first pass feature reduction: 555

second pass feature reduction with n_consensus=4: 26

small set feature selection:

selector	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
rfe_k5	0.657	0.044	0.053	0.028	0.547	0.659
elastic_selector	0.649	0.050	0.048	0.022	0.532	0.673
l1_selector	0.635	0.046	0.043	0.018	0.530	0.678
none	0.635	0.046	0.043	0.018	0.530	0.678
f_classif_k10	0.632	0.052	0.045	0.026	0.528	0.651
f_classif_k5	0.625	0.062	0.040	0.021	0.534	0.648
mi_k5	0.603	0.061	0.043	0.024	0.504	0.644

The following features are selected solely on the reduction methods - first pass as above and then second pass with consensus across n selectors, then exploratory eval with LR

n_consensus (features)	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
3 (124)	0.643	0.048	0.047	0.016	0.375	0.822
2 (273)	0.722	0.056	0.058	0.020	0.213	0.937
1 (446)	0.585	0.059	0.037	0.015	0.136	0.936

tsfresh - stratified set

Starting features: 1554

after first pass feature reduction: 1123

second pass feature reduction with n_consensus=4: 27

small set feature selection:

selector	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
l1_selector	0.696	0.058	0.062	0.023	0.524	0.717
none	0.695	0.059	0.062	0.023	0.524	0.717
elastic_selector	0.674	0.052	0.058	0.020	0.544	0.697
f_classif_k10	0.635	0.050	0.049	0.023	0.522	0.690
f_classif_k5	0.622	0.059	0.045	0.022	0.487	0.687
rfe_k5	0.621	0.053	0.042	0.019	0.497	0.670
mi_k5	0.595	0.060	0.037	0.018	0.481	0.629

The following features are selected solely on the reduction methods - first pass as above and then second pass with consensus across n selectors, then exploratory eval with LR

n_consensus (features)	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
3 (210)	0.771	0.033	0.078	0.026	0.296	0.933
2 (442)	0.815	0.042	0.092	0.032	0.212	0.968
1 (781)	0.629	0.063	0.038	0.013	0.087	0.962

Table B2. Model selection and covariate analysis full results across handcrafted, tsfresh, stratification conditions and covariate modelling:

Handcrafted - non-stratified (5 features - best set)						
model	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
NB	0.666	0.066	0.047	0.018	0.094	0.967
LR	0.652	0.064	0.045	0.018	0.565	0.652
ET	0.606	0.057	0.027	0.005	0.000	0.993
dummy	0.500	0.000	0.018	0.000	0.000	1.000

Handcrafted - stratified (5 features - best set)						
model	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
NB	0.672	0.060	0.049	0.022	0.061	0.971
LR	0.649	0.065	0.039	0.010	0.552	0.656
ET	0.540	0.051	0.023	0.003	0.000	0.998
dummy	0.500	0.000	0.018	0.000	0.000	1.000

tsfresh - non-stratified (273 features - best set)						
model	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
LR	0.722	0.056	0.058	0.020	0.213	0.937
NB	0.652	0.062	0.033	0.007	0.584	0.642
ET	0.617	0.058	0.034	0.011	0.000	1.000
dummy	0.500	0.000	0.018	0.000	0.000	1.000

tsfresh - stratified (442 features - best set)						
model	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
LR	0.815	0.042	0.092	0.032	0.212	0.968
NB	0.644	0.066	0.032	0.006	0.819	0.316
ET	0.604	0.072	0.034	0.016	0.000	1.000
dummy	0.500	0.000	0.018	0.000	0.000	1.000

SEX, ETH, BMI

model	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
LR	0.606	0.064	0.029	0.008	0.670	0.561
NB	0.593	0.067	0.028	0.008	0.000	1.000
ET	0.570	0.053	0.027	0.010	0.108	0.916
dummy	0.500	0.000	0.018	0.000	0.000	1.000

SEX, ETH, BMI + tsfresh-stratified (442)

model	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
LR	0.819	0.039	0.094	0.033	0.207	0.969
NB	0.643	0.064	0.032	0.006	0.793	0.365
ET	0.604	0.053	0.032	0.007	0.000	1.000
dummy	0.500	0.000	0.018	0.000	0.000	1.000

SEX, ETH, BMI + tsfresh-stratified (442) + SDQ

model	roc_auc_mean	roc_auc_sd	pr_auc_mean	pr_auc_std	sens_mean	spec_mean
LR	0.878	0.035	0.165	0.051	0.346	0.973
ET	0.737	0.064	0.092	0.049	0.000	1.000
NB	0.659	0.066	0.034	0.007	0.810	0.381
dummy	0.500	0.000	0.018	0.000	0.000	1.000

Covariate analysis coefficients:

SEX, ETH, BMI			SEX, ETH, BMI + tsfresh-stratified (442)		
	feature	coef		feature	coef
1	cats__SEX	0.504	1	nums__counts__fft_coefficient__attr_"abs"...	1.421
2	cats__ETH	0.267	2	nums__counts__fft_coefficient__attr_"angle"...	-1.285
3	nums__BMI	-0.019	3	nums__counts__fft_coefficient__attr_"abs"...	-1.121
			4	nums__counts__value_count__value_1_WE	-1.069
			5	nums__counts__fft_coefficient__attr_"abs"...	-1.048
			...	...	...
			28	nums__counts__fft_coefficient__attr_"angle"...	-0.756
			29	nums__counts__index_mass_quantile...	0.751
			30	cats__SEX	0.751
			31	nums__counts__fft_coefficient__attr_"abs"...	-0.738
			32	nums__counts__fft_coefficient__attr_"angle"...	-0.724

Table B3. Feature importance: The full feature set for the weekday/weekend stratification condition was assigned feature families based on interpretable groupings, aided by tsfresh documentation. The optimal selection (442 features) was analysed to see how many were retained as a proportion of the original family count. The sum absolute coefficient contribution provides each family's total contribution to the model.

Feature family	Original count	Selected count	% retained	Total abs coef contribution	% coef contribution
Frequency / spectral structure	944	354	38%	130.630	83.8%
Change, variability, and sequence structure	218	28	13%	7.129	4.6%
Distribution, extremes, and bursts	90	15	17%	6.003	3.9%
Timing, trend, and energy location	150	19	13%	5.939	3.8%
Temporal dependence and stationarity	74	13	18%	3.788	2.4%
Complexity / entropy	26	5	19%	0.905	0.6%
Other / miscellaneous	4	1	25%	0.830	0.5%
Simple activity volume / magnitude	38	7	18%	0.692	0.4%
Nonlinear dynamics / model-based	10	0	0%	0.000	0.0%
Totals	1554	442		155.915789	

Table B4. Hyperparameter tuning: The logistic regression hyperparameter search evaluated 14 combinations, covering ridge-like regularisation for stability across correlated tsfresh features and elastic-net regularisation for greater sparsity:

Search block	Model type	L1 ratio	C values	Class weight	n combinations
1	Ridge logistic regression	0	0.03, 0.1, 0.3, 1.0	"balanced", None	8
2	Elasticnet logistic regression	0.2, 0.5	0.03, 0.1, 0.3	"balanced"	6

With results sorted by AUROC:

params	mean_roc_auc	std_roc_auc	mean_pr_auc	std_pr_auc
{'clf__C': 0.1, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.0}	0.846	0.044	0.109	0.037
{'clf__C': 0.03, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.0}	0.844	0.042	0.107	0.033
{'clf__C': 0.3, 'clf__class_weight': None, 'clf__l1_ratio': 0.0}	0.844	0.043	0.108	0.038
{'clf__C': 0.3, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.0}	0.844	0.044	0.108	0.038
{'clf__C': 1.0, 'clf__class_weight': None, 'clf__l1_ratio': 0.0}	0.842	0.044	0.110	0.040
{'clf__C': 0.1, 'clf__class_weight': None, 'clf__l1_ratio': 0.0}	0.842	0.042	0.107	0.035
{'clf__C': 1.0, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.0}	0.840	0.044	0.110	0.043
{'clf__C': 0.03, 'clf__class_weight': None, 'clf__l1_ratio': 0.0}	0.831	0.039	0.106	0.033
{'clf__C': 0.3, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.2}	0.825	0.047	0.093	0.032
{'clf__C': 0.1, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.2}	0.805	0.049	0.081	0.026
{'clf__C': 0.3, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.5}	0.793	0.052	0.077	0.027
{'clf__C': 0.03, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.2}	0.762	0.048	0.063	0.018
{'clf__C': 0.1, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.5}	0.749	0.053	0.059	0.017
{'clf__C': 0.03, 'clf__class_weight': 'balanced', 'clf__l1_ratio': 0.5}	0.693	0.050	0.044	0.010

Table B5. Specificity / sensitivity threshold analysis: Exploratory threshold analysis illustrates how metrics are dependent on where the threshold for probability scores is chosen, trade-off plot below, figure B1.

Best Balanced Accuracy:

threshold	sensitivity	specificity	precision	balanced_accuracy	tp	fp	tn	fn
0.007	0.880	0.704	0.052	0.792	81	1462	3479	11
0.010	0.837	0.739	0.056	0.788	77	1289	3652	15
0.013	0.793	0.766	0.059	0.780	73	1156	3785	19
0.033	0.717	0.831	0.073	0.774	66	835	4106	26
0.020	0.750	0.797	0.064	0.774	69	1002	3939	23

Best sensitivity while specificity ≥ 0.80 :

threshold	sensitivity	specificity	precision	balanced_accuracy	tp	fp	tn	fn
0.023	0.739	0.807	0.067	0.773	68	953	3988	24
0.027	0.717	0.817	0.068	0.767	66	904	4037	26
0.030	0.717	0.824	0.071	0.771	66	869	4072	26
0.033	0.717	0.831	0.073	0.774	66	835	4106	26
0.037	0.707	0.838	0.075	0.772	65	800	4141	27

Best specificity while sensitivity ≥ 0.80 :

threshold	sensitivity	specificity	precision	balanced_accuracy	tp	fp	tn	fn
0.010	0.837	0.739	0.056	0.788	77	1289	3652	15
0.007	0.880	0.704	0.052	0.792	81	1462	3479	11
0.003	0.891	0.638	0.044	0.765	82	1787	3154	10

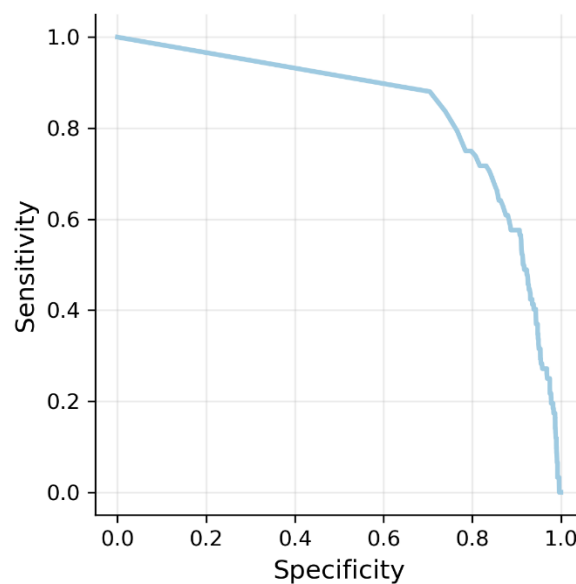


Figure B1. Plot of the trade-off between sensitivity and specificity

Table B6. Subgroup: Sex threshold analysis : For selecting a well-balanced example threshold (0.01):

group	threshold	sensitivity	specificity	precision	balanced_accuracy	tp	fp	tn	fn
All	0.500	0.193	0.970	0.104	0.581	17	146	4666	71
All	0.100	0.545	0.909	0.099	0.727	48	436	4376	40
All	0.050	0.625	0.868	0.080	0.746	55	636	4176	33
All	0.010	0.761	0.750	0.053	0.756	67	1201	3611	21
All	0.005	0.841	0.686	0.047	0.764	74	1510	3302	14
All	0.003	0.875	0.644	0.043	0.760	77	1711	3101	11